Supervisor: Selin Coban

AI supported Information Foraging

Software Construction
RWTH AACHEN UNIVERSITY

Overview

- Task Introduction
- Possible Improvements to Status Quo
- Live Demo
- Software Structure
  - AI Support
  - Knowledge Manager
- Outlook (not Microsoft)

Task Introduction

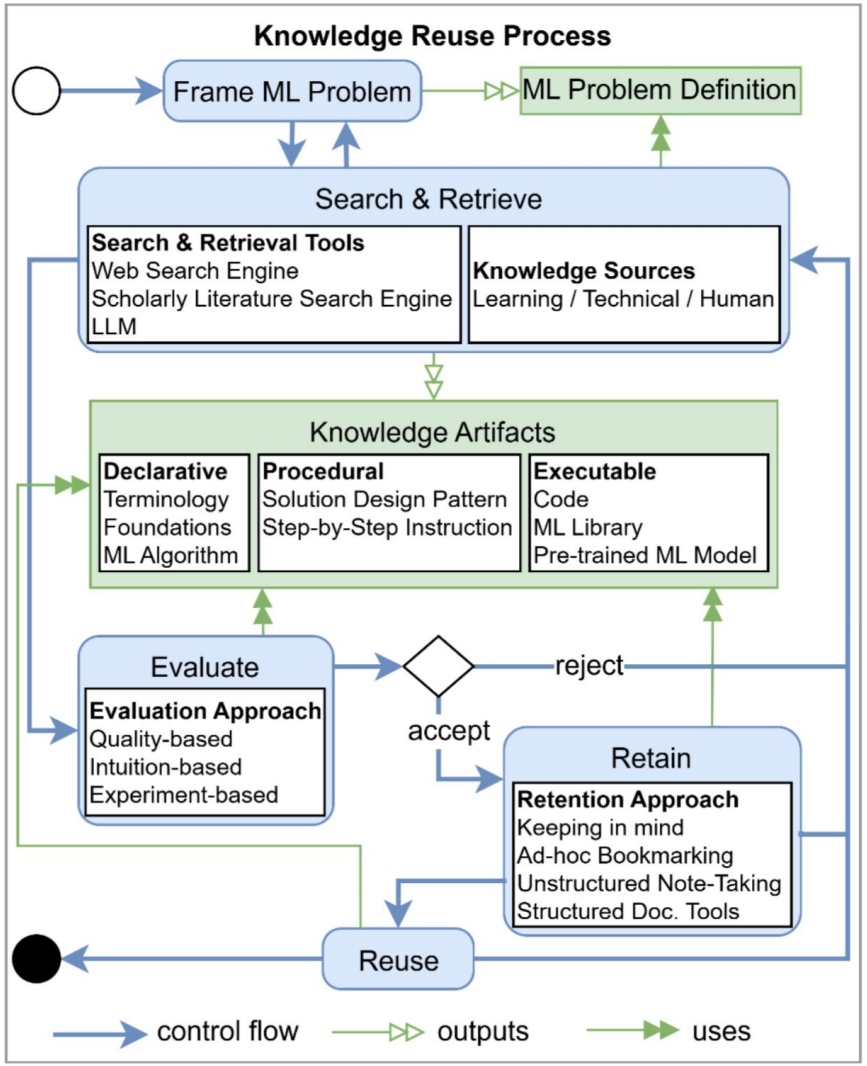
What Is Information Foraging?

- Process of gathering useful information
- Originates from Optimal Foraging Theory
- Our case:
  - How ML-Researchers do it
- Task: Improve this process
  - First: Understand domain

Domain: Information Foraging Process

- Study with 18 participants from industry/university

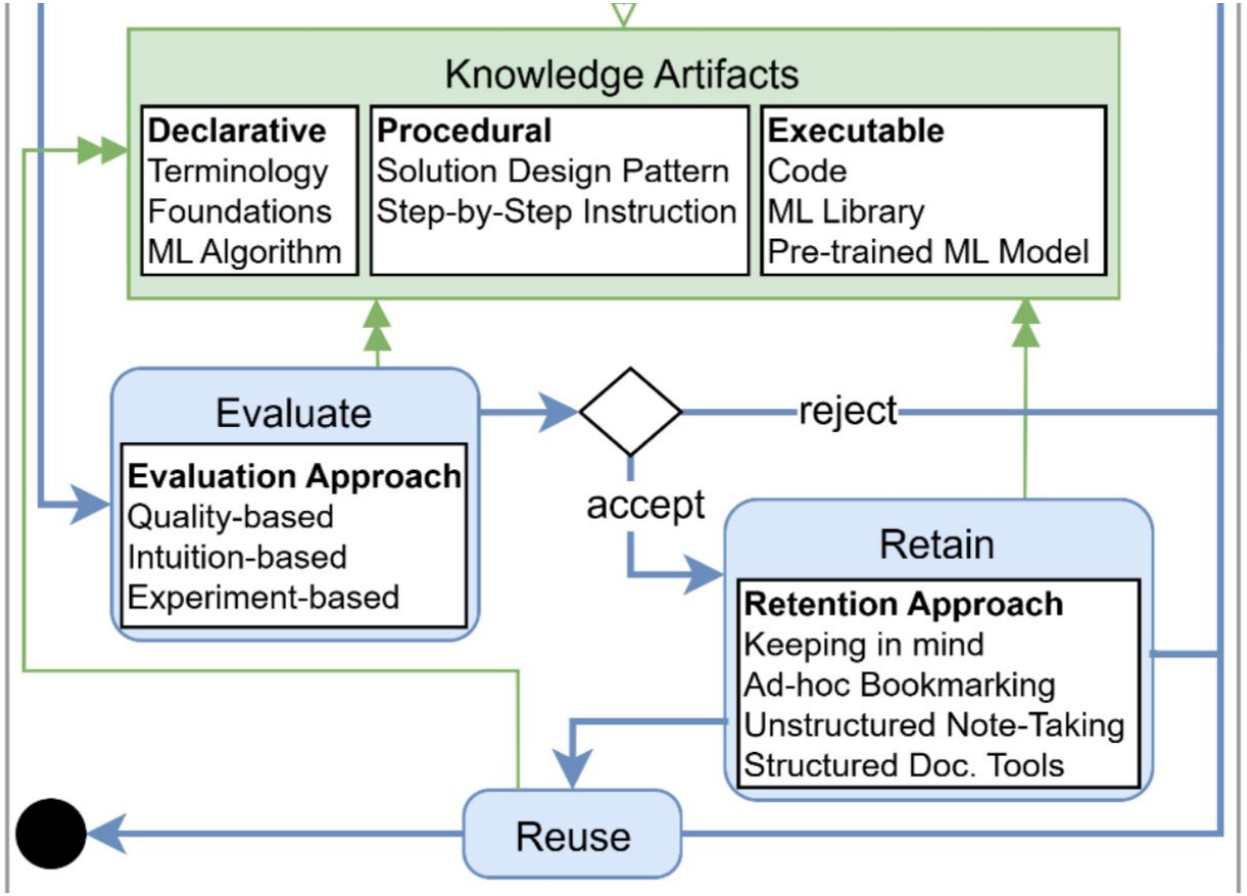
- How is information gathered?
- When is it stored?
- How is it reaccessed?



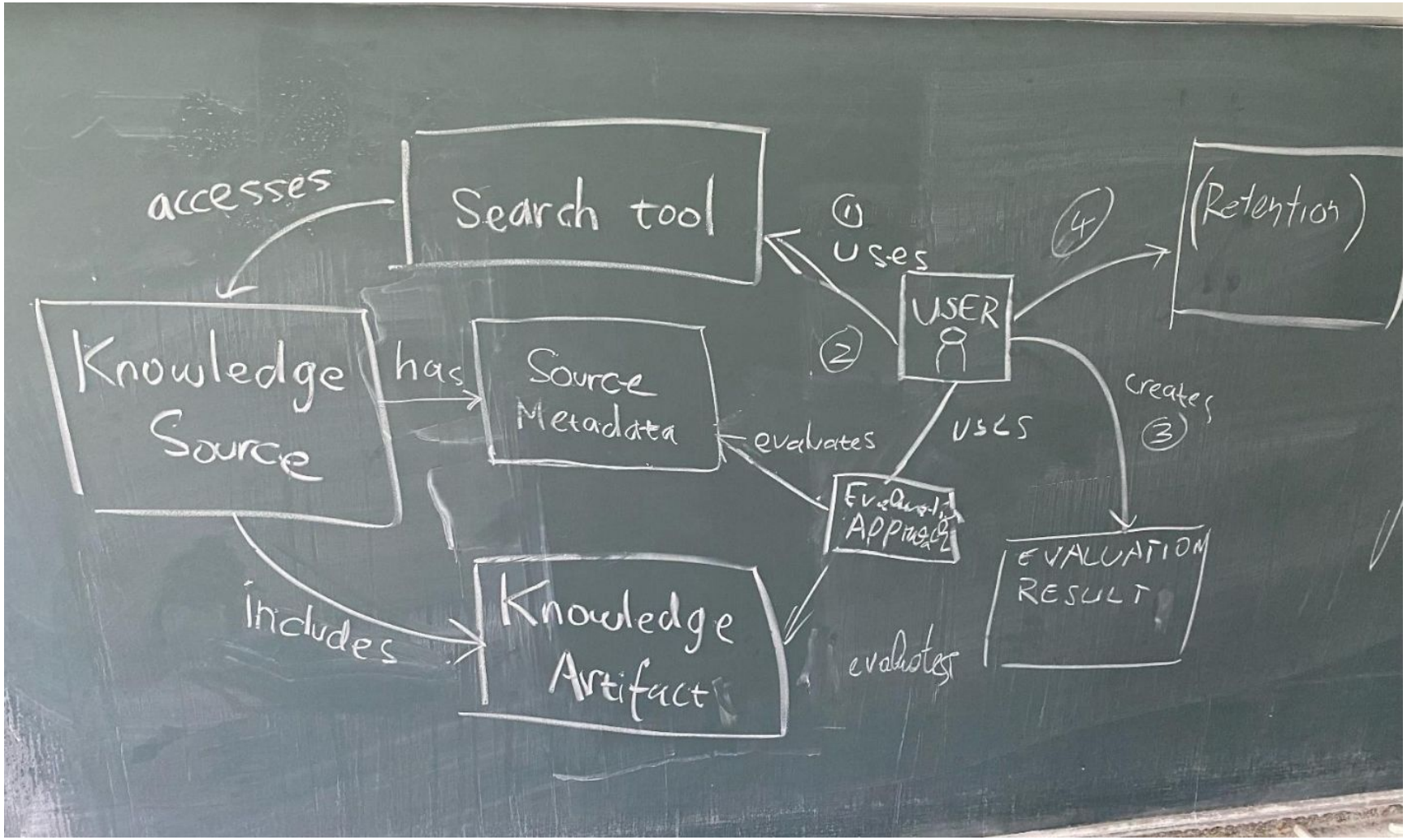
[6]

What, Why and Where Is It Stored?

1. Each Knowledge Source (KS) contains Knowledge Artifacts (KA)
   a. Code
   b. An algorithm
   c. Clear terminology, etc…
2. Knowledge Sources can be evaluated
   a. Usability
   b. Credibility
   c. Relevance
3. Retained
   a. Keeping in mind
   b. Bookmarking
   c. Notes
   d. Zotero, etc…



Our Domain Story



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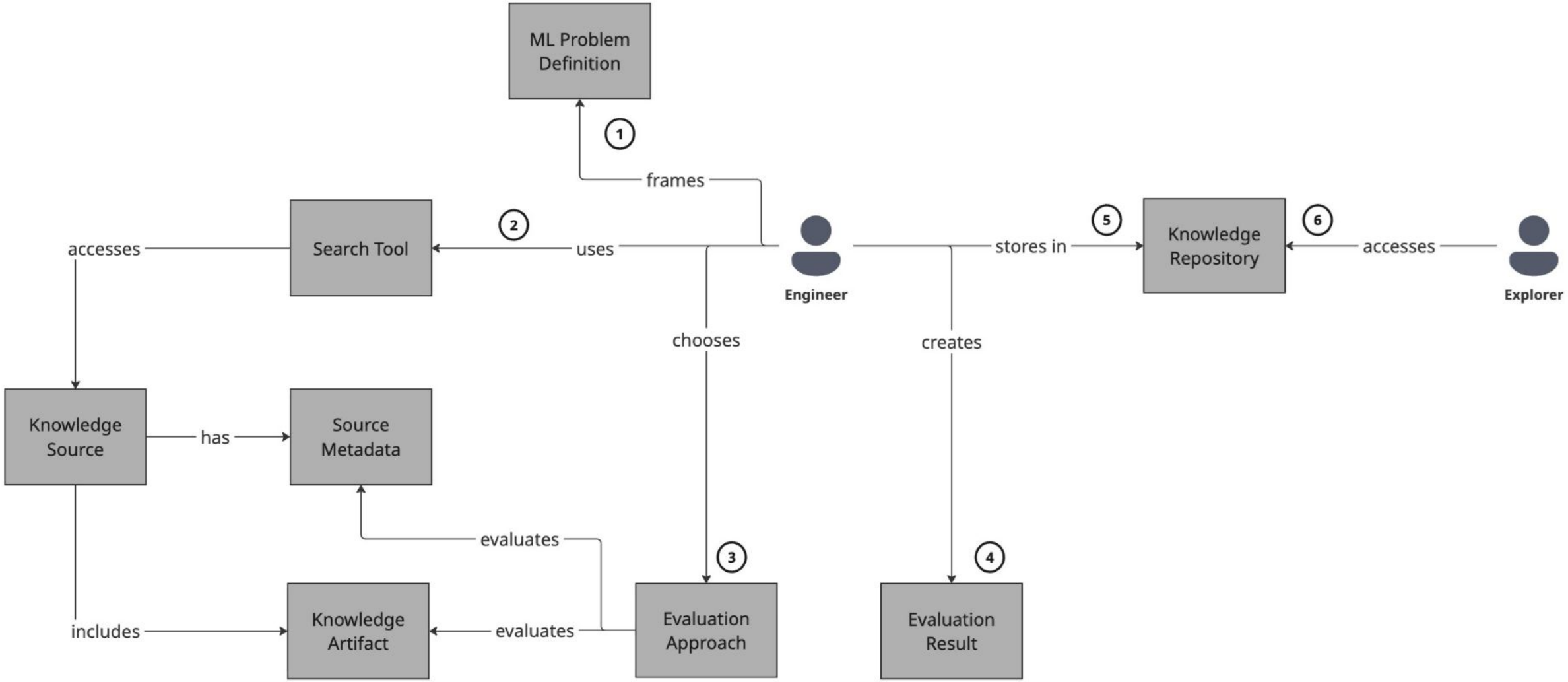
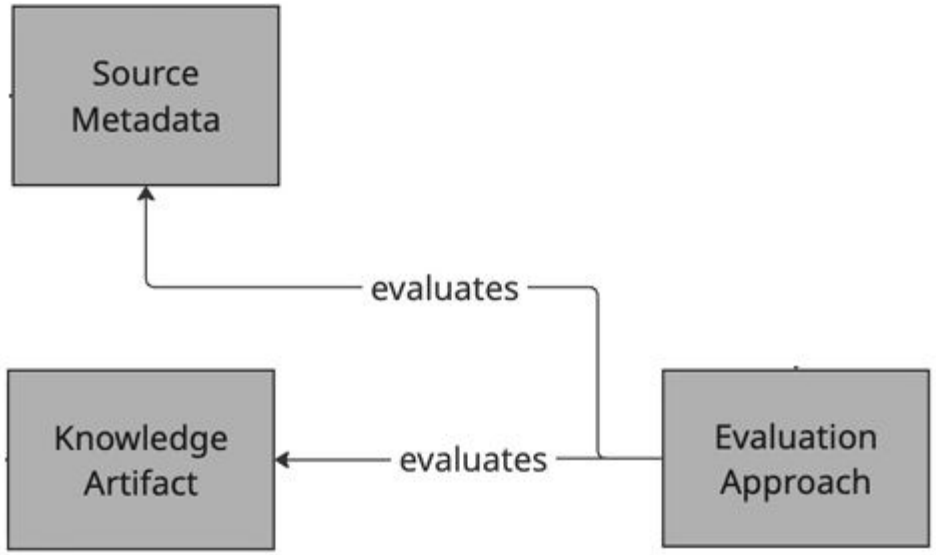


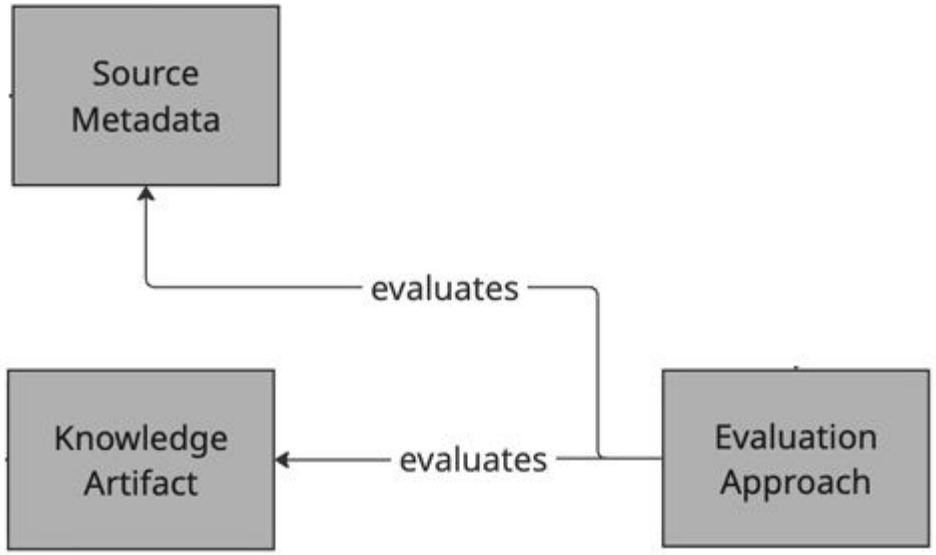
Possible Improvements

What Can We Improve?



What Can We Improve?

- Filter out low quality KS
- Extracting KAs from KS
- Understand KAs faster

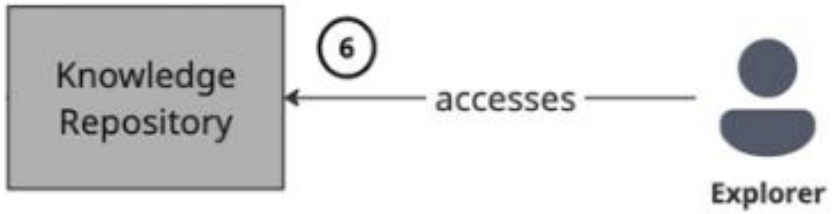


What Can We Improve?

What Can We Improve?

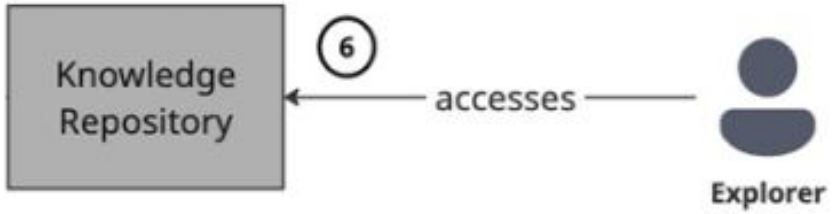
- Save evaluation results of KS
- Group KS by project

What Can We Improve?



What Can We Improve?

- Make information easily findable
- Retrace where KAs were used



What Can We Improve?

- Filter out low quality KS
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- Understand KAs faster

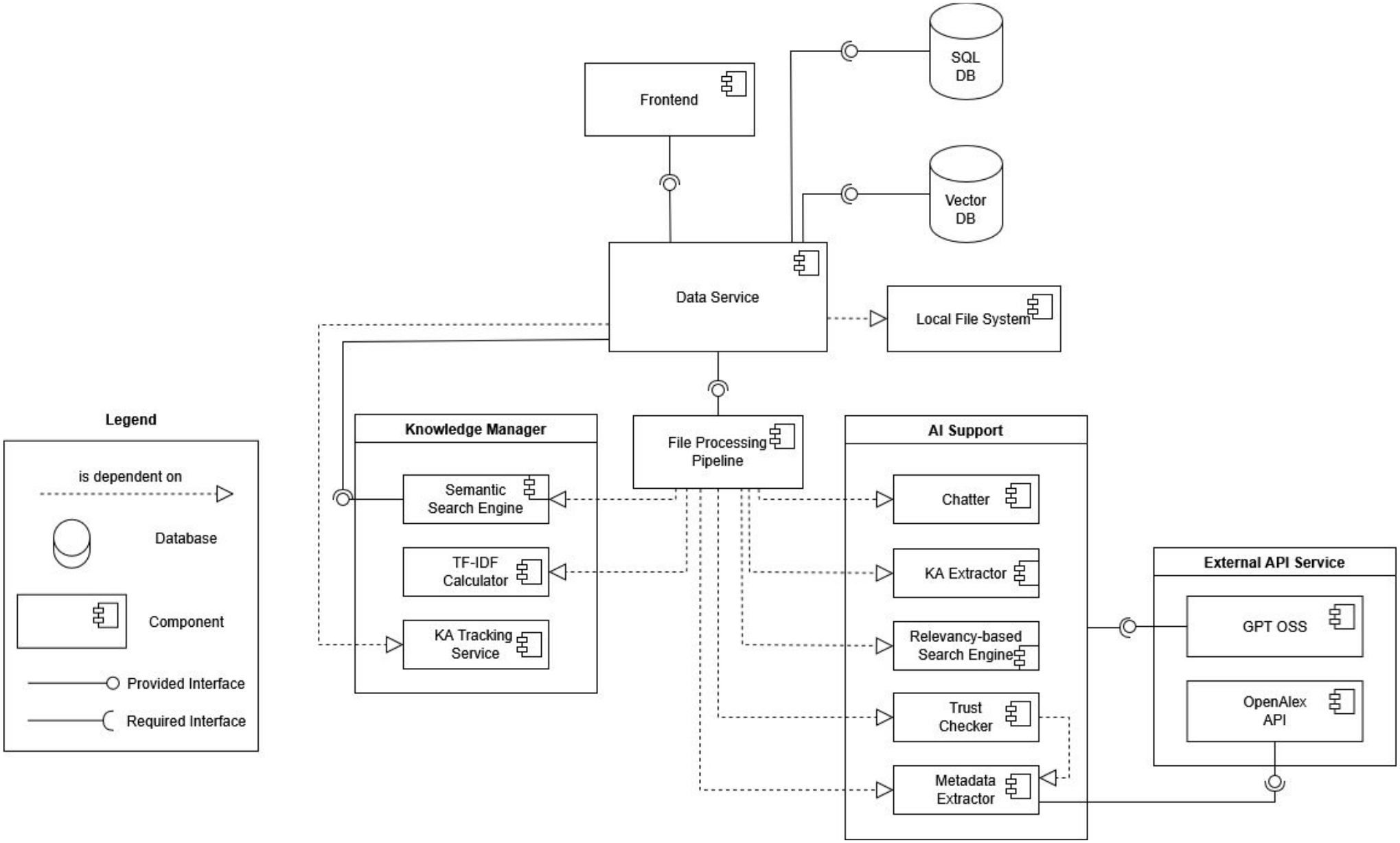
→ AI Support

- Save evaluation results of KS
- Group KS by project
- Make information easily findable
- Retrace where KAs were used

→ Knowledge Source Management

Live Demo

Software Structure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Vector DB
Data Service
Local File System
Legend
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Component
Provided Interface
Required Interface
Knowledge Manager
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TF-IDF Calculator
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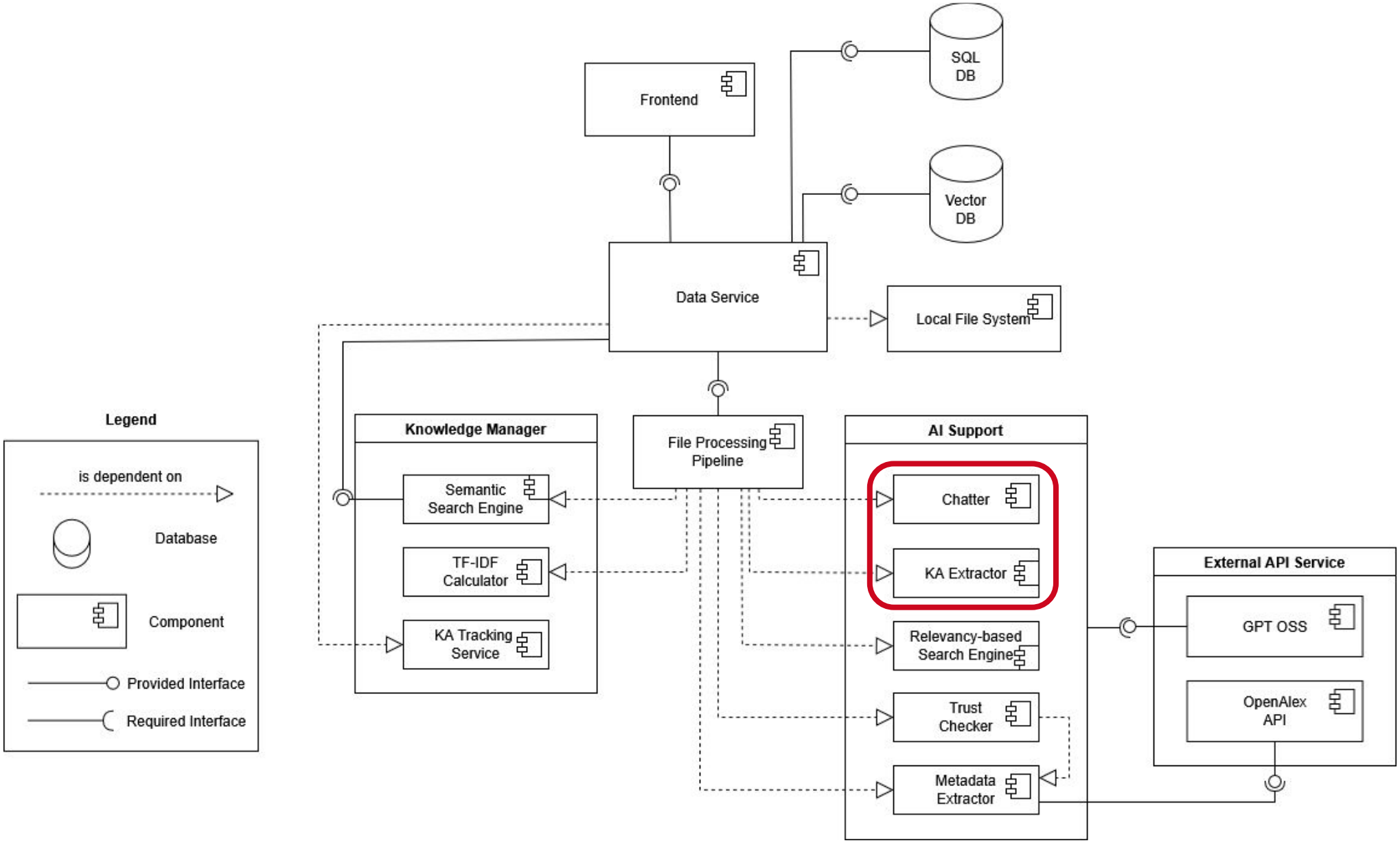
File Processing Pipeline
AI Support
Chatter
KA Extractor
Relevancy-based Search Engine
Trust Checker
Metadata Extractor
External API Service
GPT OSS
OpenAlex API

[2]

AI Support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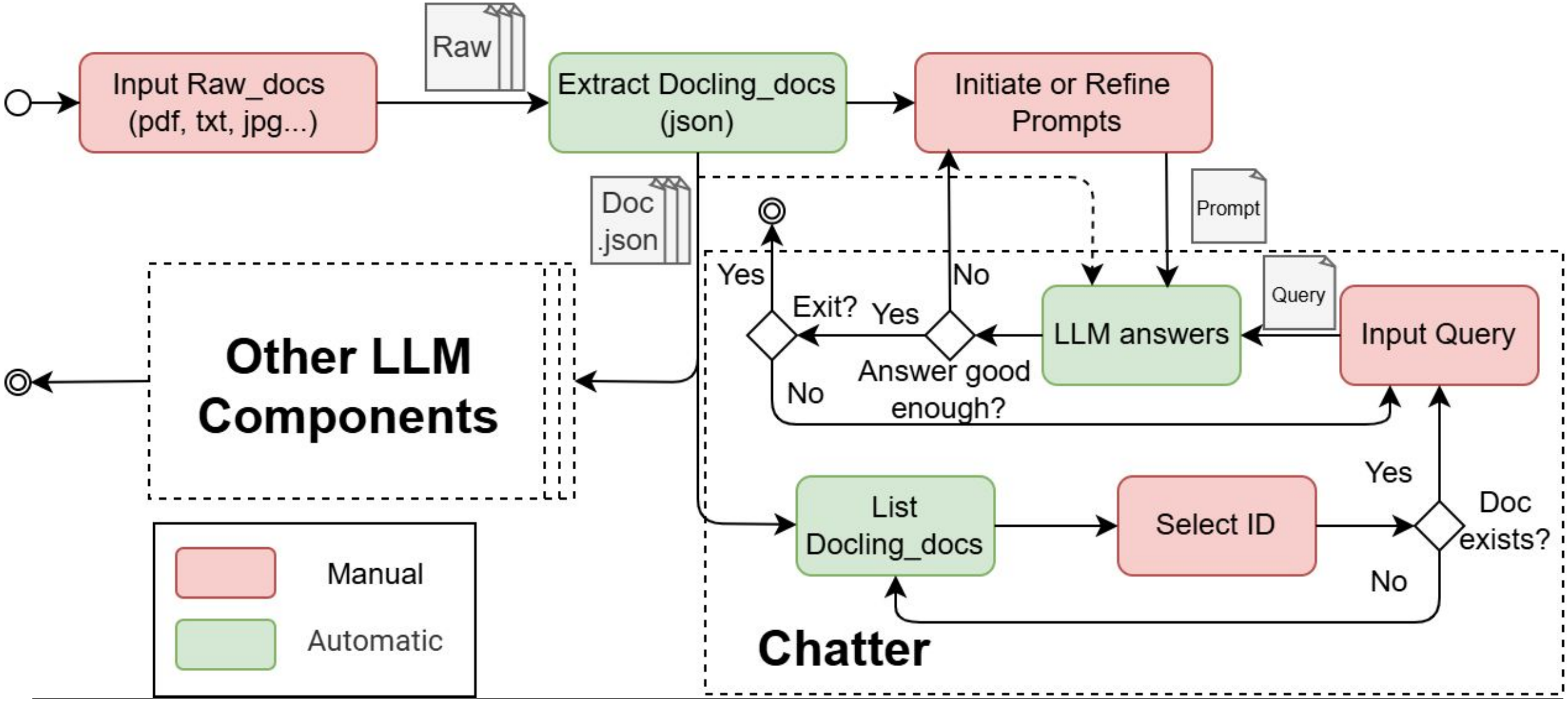
[2]

The Rules we Follow to Create LLM Prompts

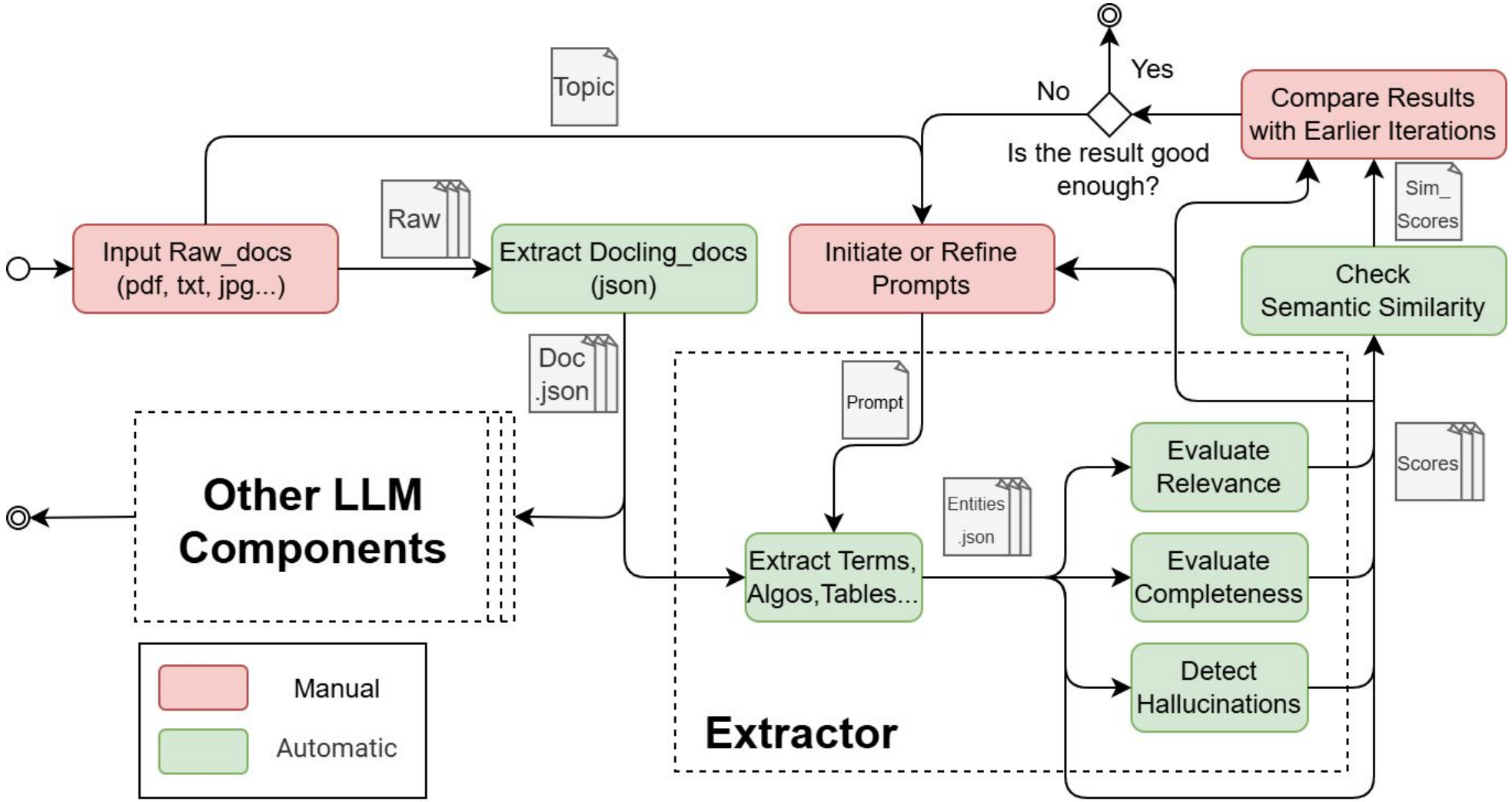
- Persona Pattern
  - Ask LLM to act as a specialist (AI Research Assistant, Academic Reviewer …)
- Template prompt pattern and zero-shot prompting strategy
  - Provides a predefined structure, but no prior examples
- Iterative prompt refinement
  - Refine the prompts iteratively
- Self-assessment
  - Let LLM provide a relevance score and a justification for the score
  - Let LLM provide a completeness score and the missing item list
  - Let LLM detect hallucinations and provide the ambiguous item list

[5]

Chatter Workflow



Extractor Workflow



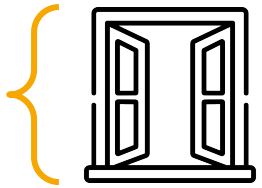
How to Deal with Large Files and Long Documents?

MapReduce

[1]

Smart Chunking

Sometimes, the document we want to process is too large for the LLM.



LLM Context Window

Smart Chunking

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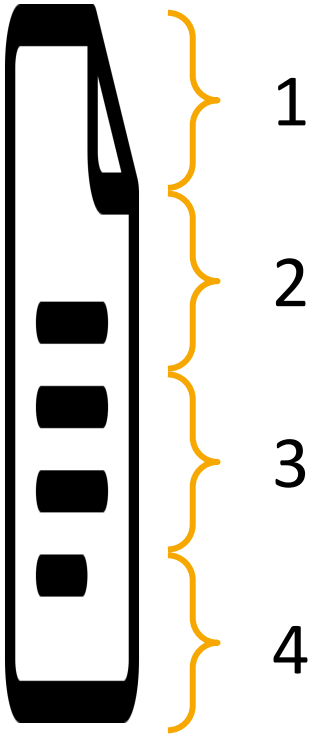
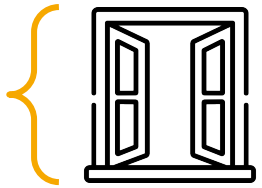


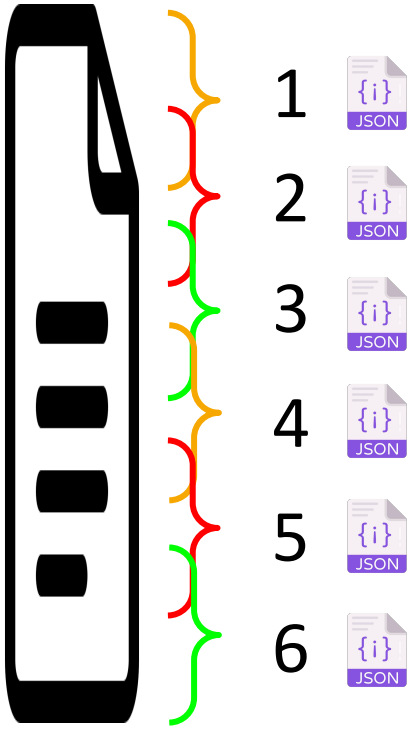
Our document



LLM Context Window

Smart Chunking

Map:



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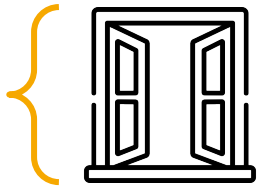

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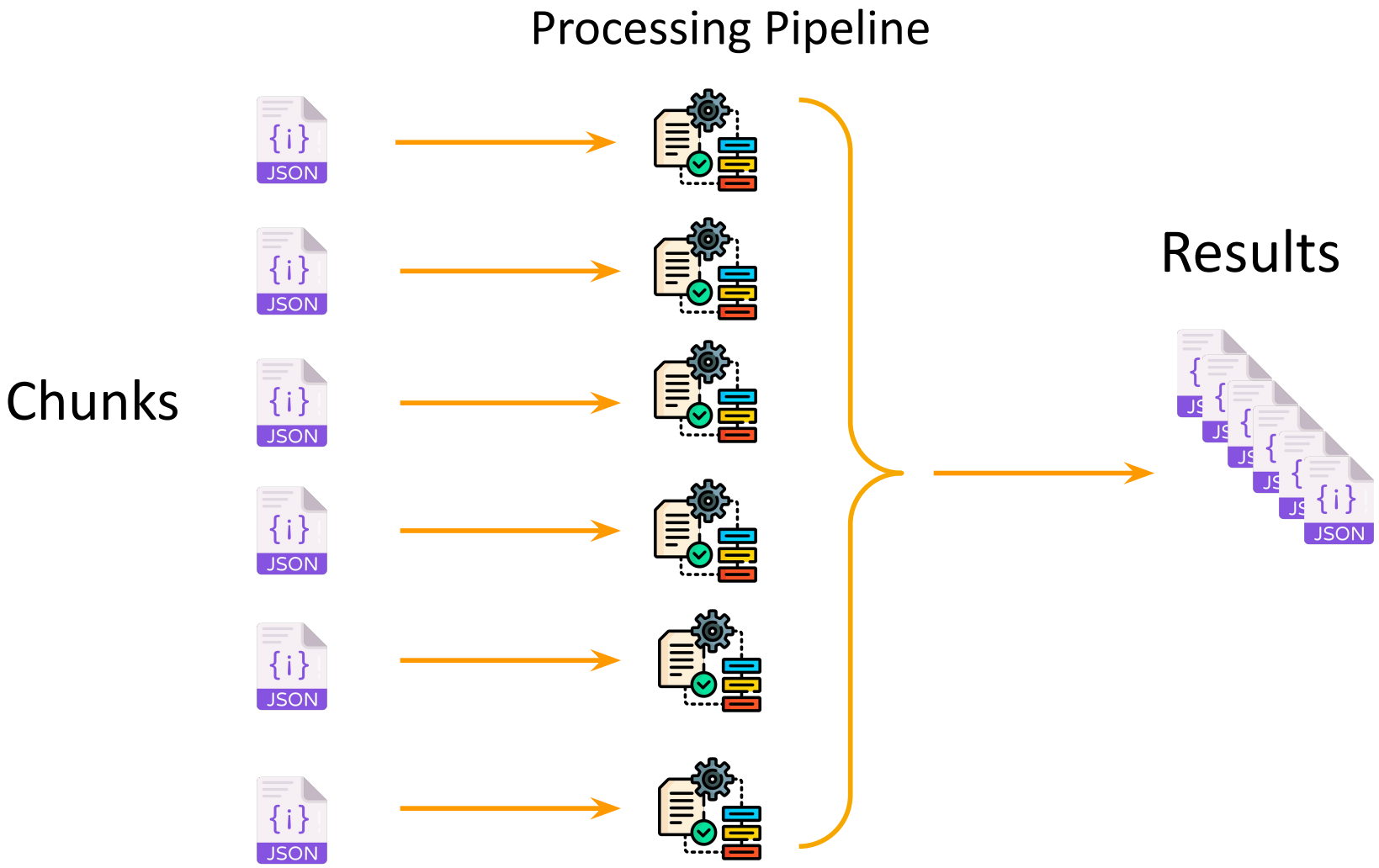

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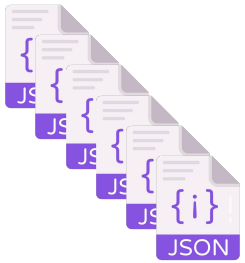
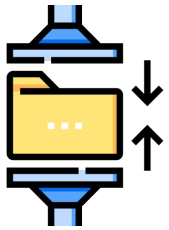
Smart Chunking

Reduce:

Results of Chunks

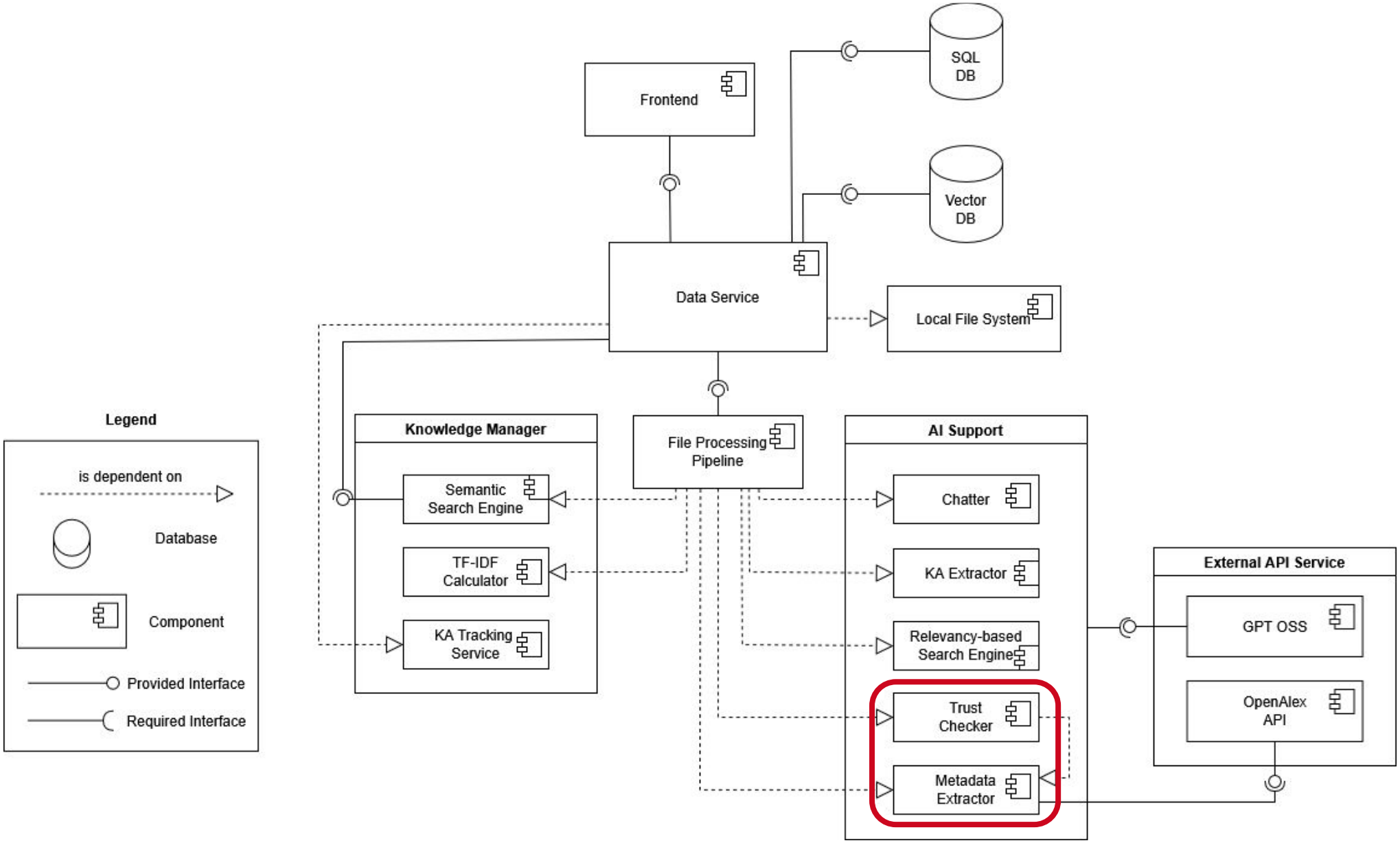


Deduplication



Reduce

Clean Knowledge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[2]

Metadata Extraction and Trust Checker

LLM-Assisted Metadata Extraction

Select only the Title Block, Abstract, and Introduction

1. Use an LLM to identify bibliographic fields (title, authors, year, venue, DOI)
   - Works even when metadata is implicit, reordered, or unstructured

2. Scholarly API verification and enrichment (OpenAlex API)
   - If a DOI is found → exact lookup
   - Otherwise → title-based fuzzy matching

   Enriches metadata (citation count, canonical URL, venue)

3. Consistency and normalization (synchronize DOI and URL)
   - Makes the DOI and the URL consistent
   - If no canonical URL is found, generate a Google Scholar search link

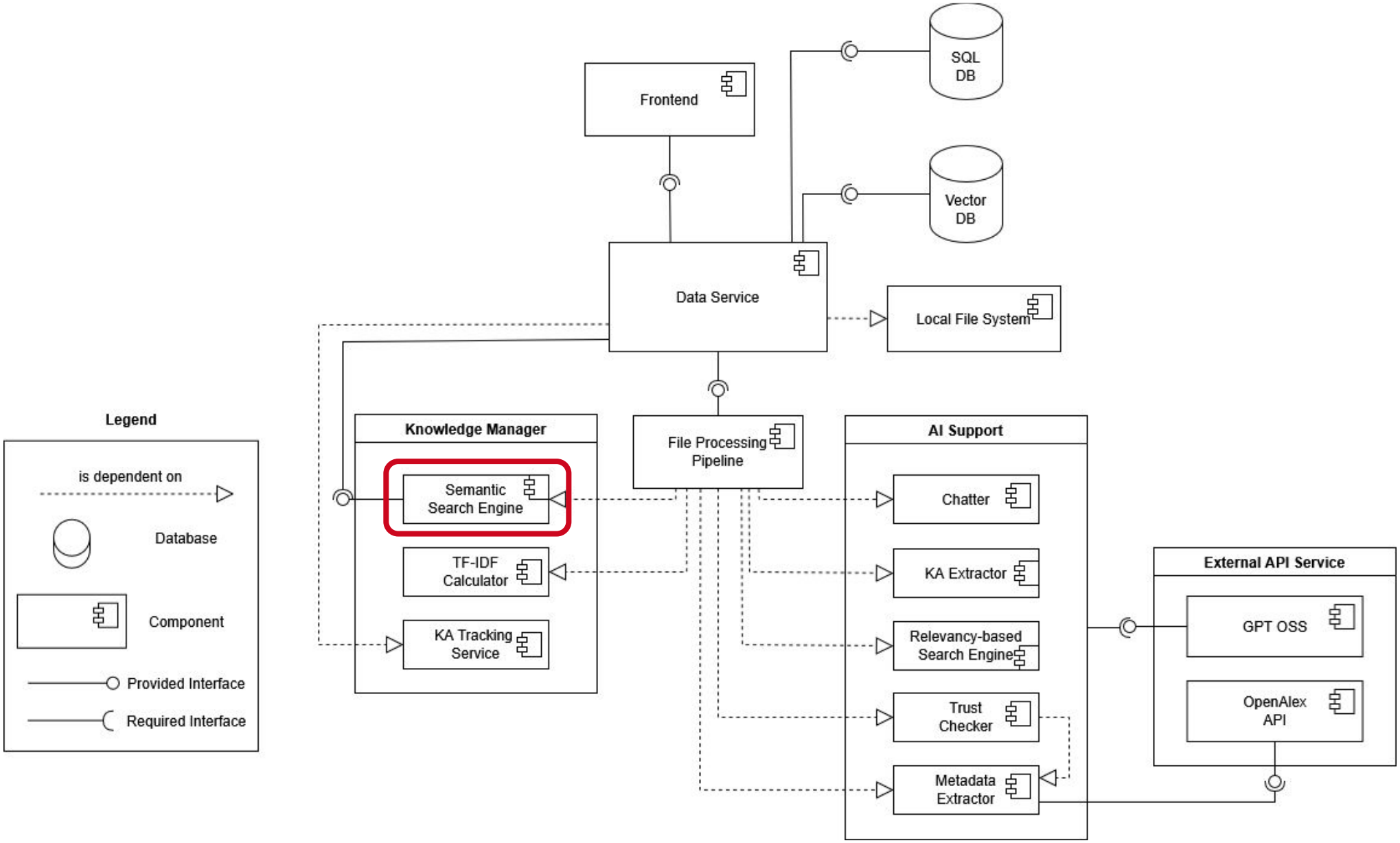
Trust Checker

Extracted Metadata + Knowledge Source Content

gpt-oss API hosted on HPC Aachen

[2]

Knowledge Manager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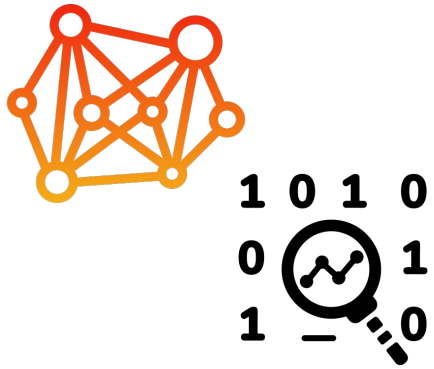
[2]

Search Types

1. Traditional Keyword Search

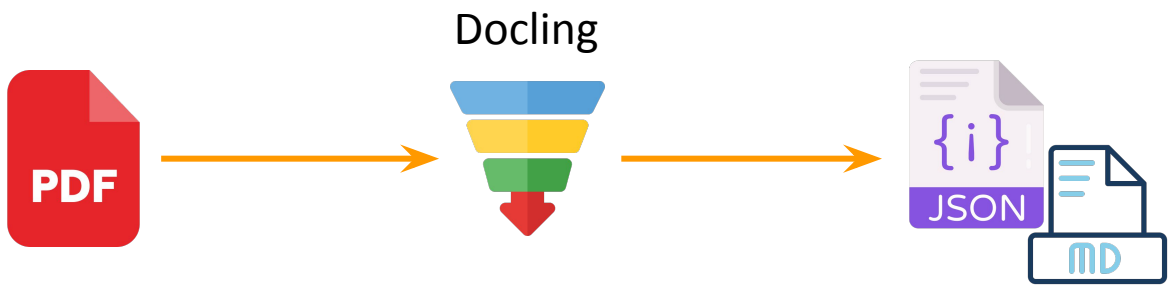
2. Semantic Search



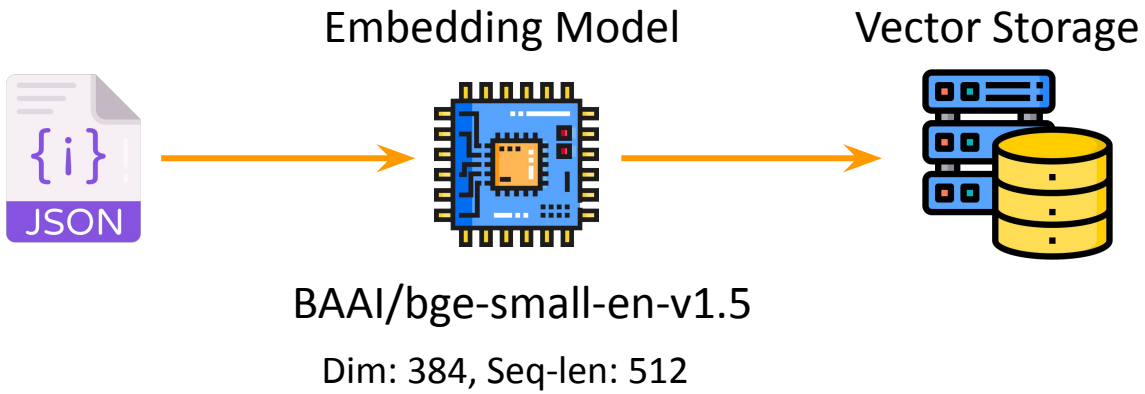
Semantic Search – Retrieval-Augmented Generation (RAG)

Recap:

1. Document Processing



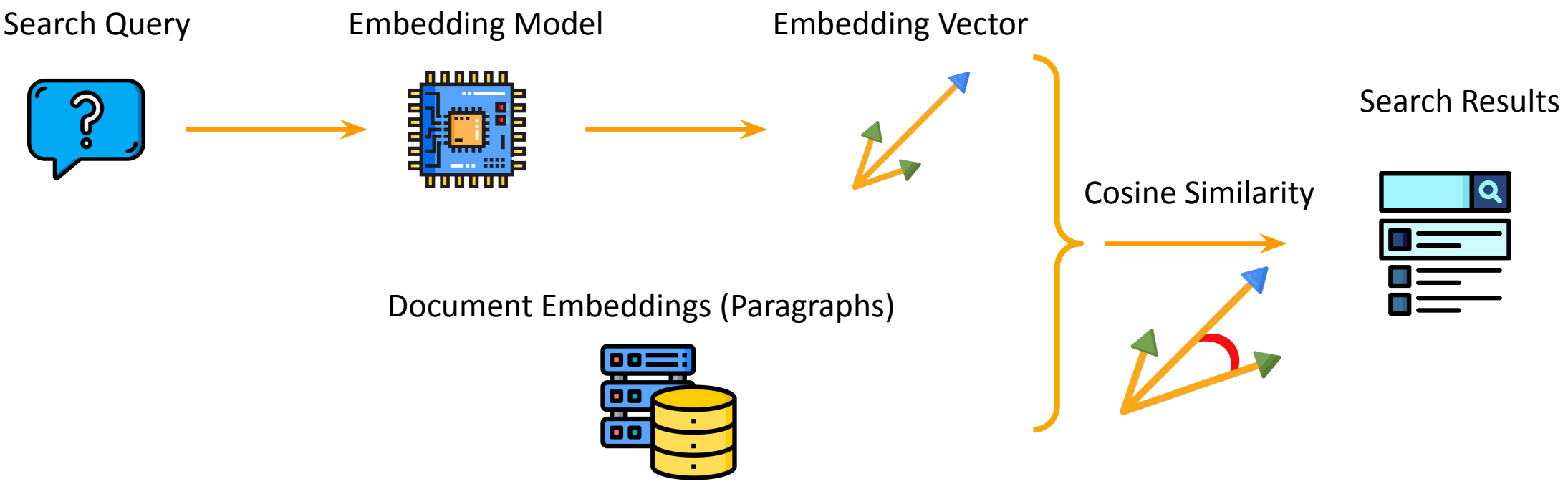
2. Embed the JSONs (paragraphs) into Vector Representations



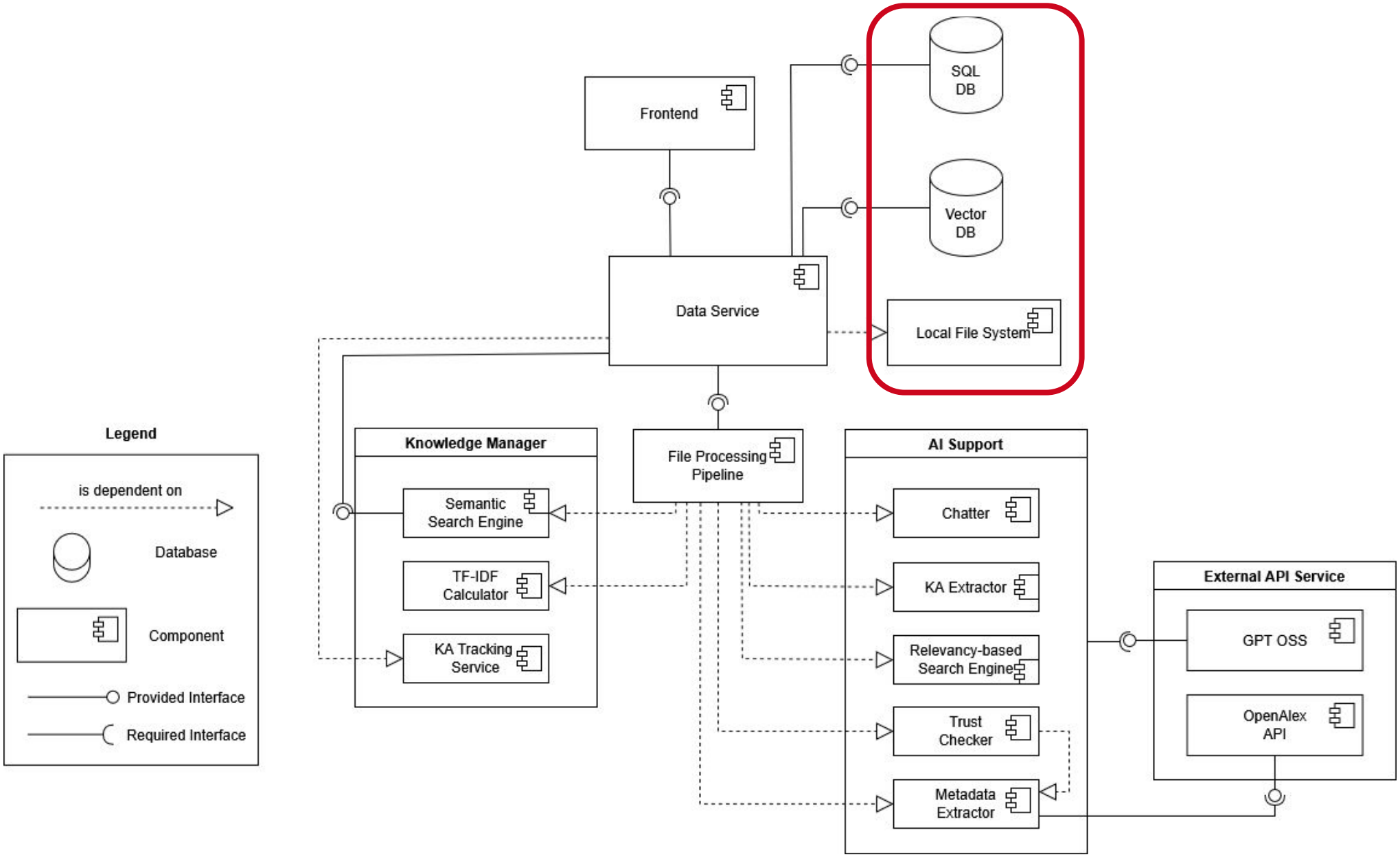
[3]

Semantic Search – Retrieval-Augmented Generation (RAG)

3. Embed the search query and compare to the document embeddings using cosine-similarity



Storage Implementation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[2]

The Data Layer (Storage and Retrieval)

- Vector Database: ChromaDB
  - Local-first, open-source, python-native.
  - Enables semantic search (finding concepts, not just keywords)

- Relational Metadata: SQLite
  - Serverless, robust, portable (inspired by Zotero)

- File Storage: Proprietary File Service
  - Raw files (PDF, DOCX, HTML, JPG, etc.)
  - Parsed JSON

[4]

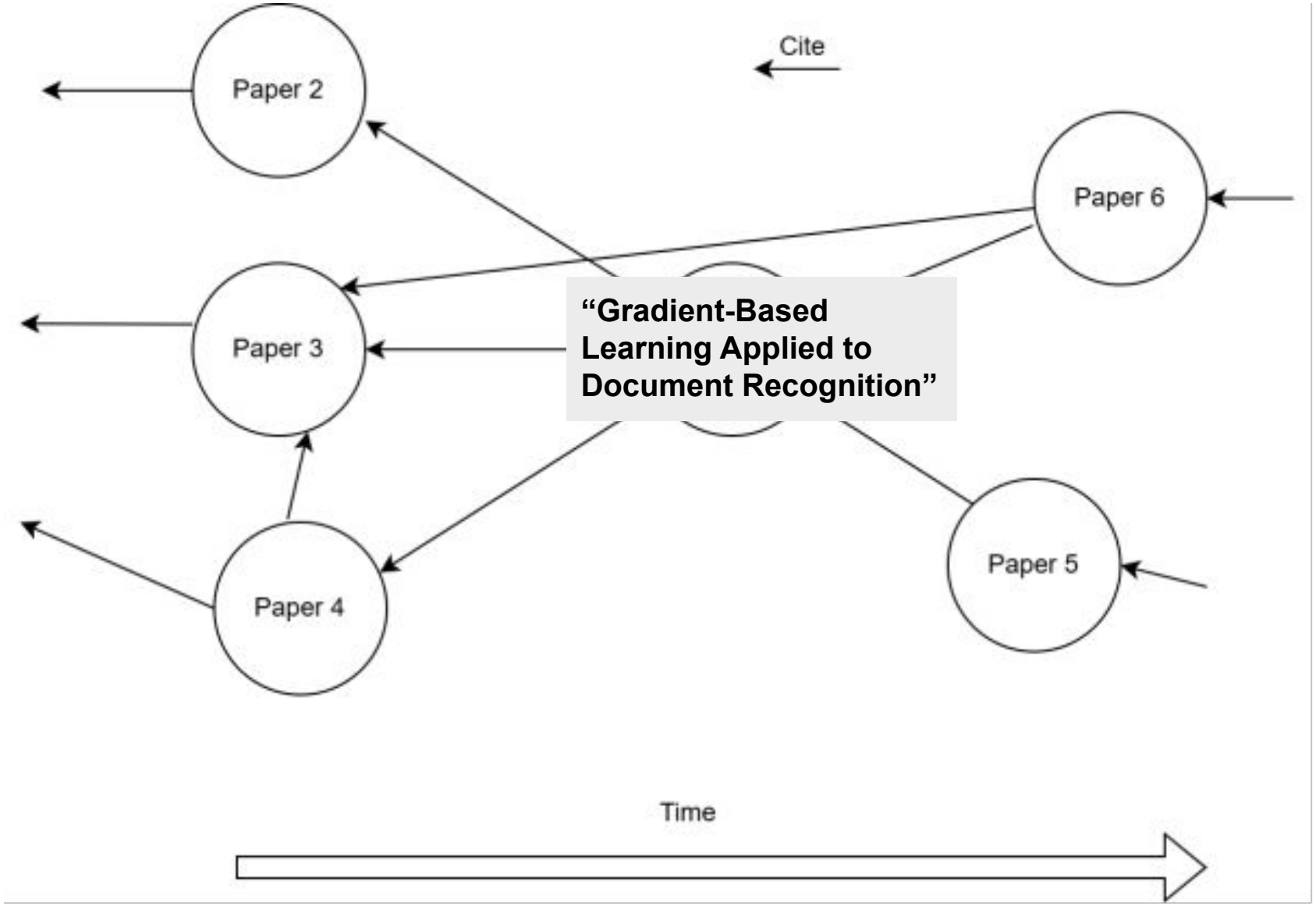
Outlook

Outlook

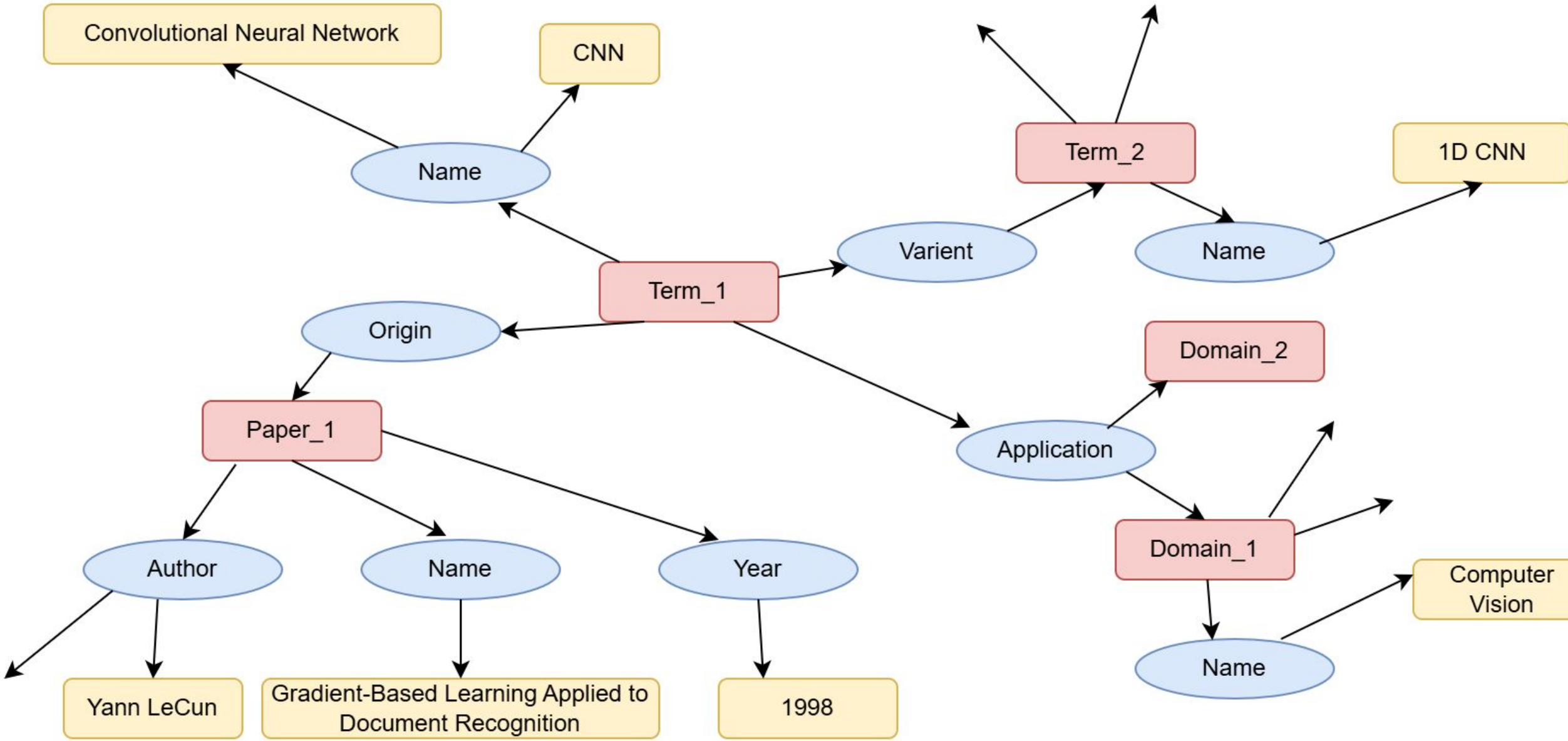
- Implementing multiple Users (collaborative mode)
- Use cloud to store all data
- Importing blog articles should be possible

AI Supported Semantic Knowledge Platform

for Citation-Driven Information Foraging



“Gradient-Based Learning Applied to Document Recognition”


Convolutional Neural Network
CNN
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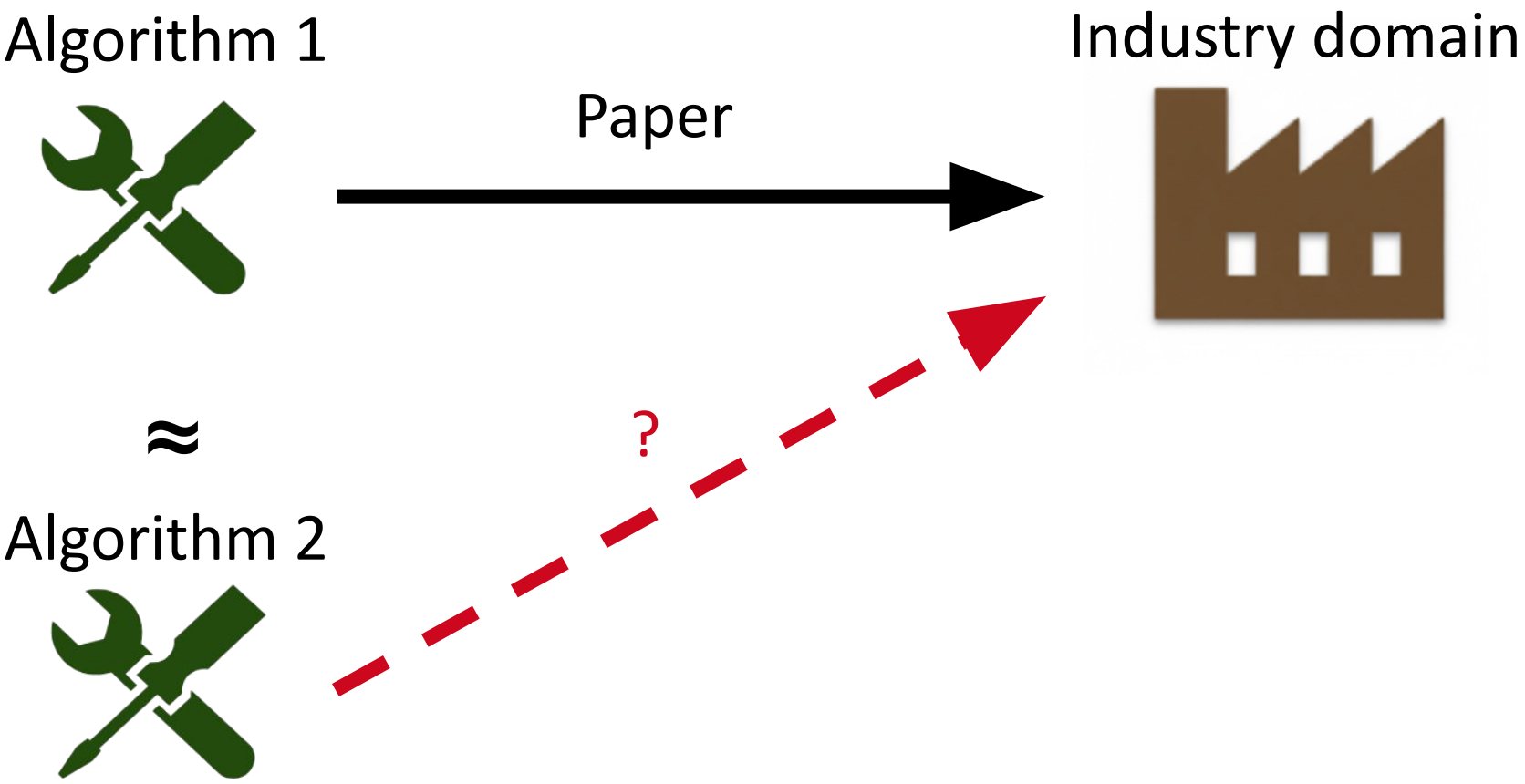
Term_2
1D CNN
Varient
Name
Term_1
Origin
Domain_2
Paper_1
Application
Author
Name
Year
Domain_1
Computer Vision
Name
Yann LeCun
Gradient-Based Learning Applied to Document Recognition
1998

- **Term (self-reference)**

  Convolutional Neural Network (CNN)
- **Origin (where it was introduced)**

  LeCun et al., *"Gradient-Based Learning Applied to Document Recognition"* (1998)
- **Further improvements (key papers that developed it)**
  - Krizhevsky et al., *AlexNet* (2012) — deep CNNs + GPUs
  - Simonyan & Zisserman, *VGG* (2014) — very deep, uniform architectures
  - He et al., *ResNet* (2015) — residual connections for extremely deep CNNs
- **Variants (how many / what kinds)**
  - 1D CNN (time series, signals)
  - 2D CNN (images)
  - 3D CNN (video, volumetric data)
  - Dilated / Atrous CNN
  - Depthwise-separable CNN (e.g. MobileNet)
- **Application domains (where it is widely used)**
  - Computer vision (image classification, detection, segmentation)
  - Medical imaging
  - Autonomous driving
  - Remote sensing
  - Industrial inspection


Algorithm 1
Industry domain
Paper
≈
?
Algorithm 2

References

[1] Dean & Ghemawat (2008). MapReduce: Simplified Data Processing on Large Clusters. Communications of the ACM, 51(1), 107–113.

[2] OpenAI et al. (2025). gpt-oss-120b & gpt-oss-20b Model Card. arXiv:2508.10925.

[3] BAAI (2023). BAAI/bge-small-en-v1.5 embedding model. Hugging Face model card. Available at https://huggingface.co/BAAI/bge-small-en-v1.5

[4] ChromaDB. (2022–2026). Chroma: The open-source vector database for AI applications (Version 1.4.1) [Software]. Apache-2.0 License. https://github.com/chroma-core/chroma

[5] Alex Mattukat. *Can ChatGPT Generate Realistic Synthetic System Requirement Specifications? A Case Study*.

[6] Selin's original Kickstarter Presentation for the lab

Thank you!